

Unified Foundation Model Framework for Bidirectional Generation of Medical Images and Findings from Mixed Lung CT and Cytological Data

Atsushi Teramoto^{1(✉)}, Maiko Nagao¹, Kaito Urata¹, Masashi Kondo²,
Kazuyoshi Imaizumi² Yuka Kiriya², Tetsuya Tsukamoto^{2,3}, and Hiroshi Fujita⁴

¹ Meijo University, Nagoya Aichi 468-8502, Japan

² Fujita Health University, Toyoake Aichi 470-1192, Japan

³ Memorial Sloan Kettering Cancer Center, New York, NY 10065, USA

⁴ Gifu University, Gifu 501-1194, Japan
teramoto@meijo-u.ac.jp

Abstract. We propose a unified foundation model framework for bidirectional generation of lung images and findings, focusing on mixed datasets comprising CT and cytological images. In this study, we constructed a dataset of 854 paired samples, where physicians provided detailed image findings for clinically significant CT and cytological images from 120 patients with suspected lung cancer. We employed three Image-to-Text model and the Stable Diffusion v2 model for Text-to-Image model. All models were fine-tuned using our dedicated dataset containing both CT and pathology images. The generated findings achieved BLEU4 of 0.693, CIDEr of 5.846, showing strong consistency with expert annotations. The Text-to-Image model produced realistic images that clearly distinguished between CT and cytology modalities, achieving comparable performance to models trained on single modalities. These results indicate the significant potential of foundation models for unified interpretation and generation across heterogeneous medical modalities in real clinical scenarios.

Keywords: Vision-Language Foundation Models, Computed Tomography, Cytological Image, Image Generation, Image Captioning

1 Introduction

Lung cancer diagnosis involves several steps, from detecting lesions in radiological images to confirming the disease with pathological images. CT and cytological images are commonly collected in this process. Many AI methods have been proposed for lung image analysis, and image generation techniques have also been explored to address data imbalance. However, conventional models are usually designed for a single modality, making it difficult to integrate multiple image types. Recently, large vision and language models [1,2] have enabled foundation models that process different modalities. In this study, we developed a unified framework for Image-to-Text and Text-to-Image tasks using lung CT and cytological images.

2 Methods

We collected data from 120 patients who underwent both CT and cytological examinations for suspected lung cancer. For CT, we focused on lung nodules, and for pathology, we used Papanicolaou-stained cytological images from biopsy samples. From these images, diagnostically useful ones were selected, and image findings were written in English by pathologists and pulmonologists. As a result, we constructed a dataset of 854 image–text pairs. We adopted BLIP [1], BLIP-2, and GiT as Image-to-Text models, and Stable Diffusion v2 [2] as the Text-to-Image model. These models were fine-tuned using our dataset containing both CT and cytological images.

3 Results

Figure 1 shows examples of image findings generated by the Image-to-Text and Text-to-Image models. Both tasks were performed on CT and cytological images. To evaluate the similarity of the generated captions, BLEU4, CIDEr, and SPICE scores were calculated, resulting in 0.693, 5.846, and 0.731, respectively.

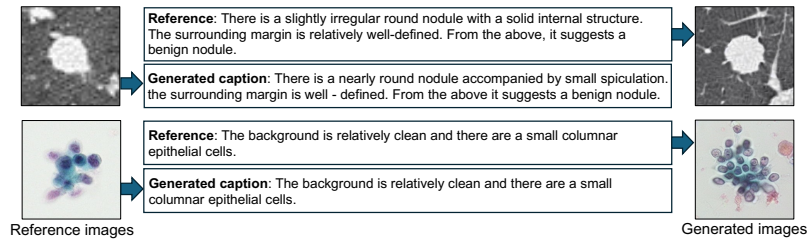


Fig. 1. Generation results

4 Conclusion

In this study, we developed a unified foundation model for bidirectional generation using both lung CT and cytological images. Our framework successfully handled two different types of medical images and produced high-quality findings and synthetic images. These results suggest that large vision-language models are effective for both understanding and generating medical data across multiple modalities.

References

1. Li, J., Li, D., Xiong, C., Hoi, S.C.H.: BLIP: Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation. In: Proceedings of the 39th International Conference on Machine Learning (ICML), pp. 12888-12900 (2022)
2. Rombach, R., Blattmann, A., Lorenz, D., Esser, P., Ommer, B.: High-Resolution Image Synthesis with Latent Diffusion Models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 10684-10695 (2022)